

Problem

Assuming $R = 2 \text{ k}\Omega$, design a parallel RLC circuit that has the characteristic equation

$$s^2 + 100s + 10^6 = 0.$$

Step-by-step solution

Step 1 of 3

Consider the general characteristic equation of parallel RLC circuit.

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0 \dots\dots (1)$$

Consider the resistance R is equal to $2 \text{ k}\Omega$.

Consider the characteristic equation of parallel RLC circuit as,

$$s^2 + 100s + 10^6 = 0 \dots\dots (2)$$

Step 2 of 3

Compare equation 1 with equation 2.

Compare constant terms.

$$\frac{1}{LC} = 10^6 \dots\dots (3)$$

Compare s terms.

$$\frac{1}{RC} = 100 \dots\dots (4)$$

Determine the value of capacitor.

Replace $2 \text{ k}\Omega$ for R in equation 4.

$$\frac{1}{RC} = 100$$

$$\frac{1}{(2 \times 10^3)C} = 100$$

$$C = \frac{1}{(2 \times 10^3)100}$$

$$C = 5 \text{ }\mu\text{F}$$

Therefore, the capacitance value is equal to $5 \text{ }\mu\text{F}$.

Step 3 of 3

Determine the value of the inductor.

Replace $5\text{ }\mu\text{F}$ for C in equation 3.

$$\frac{1}{L(5 \times 10^{-6})} = 10^6$$

$$L = \frac{1}{10^6(5 \times 10^{-6})}$$

$$L = 0.2\text{ H}$$

Therefore, the value of the inductor is 0.2 H.

Draw the circuit as shown in Figure 8.1 (b) by replacing $2\text{ k}\Omega$ for R , $5\text{ }\mu\text{F}$ for C , and 0.2 H for L .